

Psycholinguistic Analyses in Software Engineering Text: A Systematic Mapping Study

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Abstract

Context: A deeper understanding of human factors in software engineering (SE) is essential for improving team collaboration, decision-making, and productivity. Communication channels like code reviews and chats provide insights into developers' psychological and emotional states. While large language models excel at text analysis, they often lack transparency and precision. Psycholinguistic tools like Linguistic Inquiry and Word Count (LIWC) offer clearer, interpretable insights into cognitive and emotional processes exhibited in text. Despite its wide use in SE research, no comprehensive mapping study of LIWC's use has been conducted.

Objective: We examine the importance of psycholinguistic tools, particularly LIWC, and provide a thorough analysis of its current and potential future applications in SE research.

Methods: We conducted a systematic mapping study of six prominent databases, identifying 43 SE-related papers using LIWC. Our analysis focuses

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on five research questions: *RQ1. How was LIWC employed in SE studies, and for what purposes?*, *RQ2. What datasets were analyzed using LIWC?*, *RQ3: What Behavioral Software Engineering (BSE) concepts were studied using LIWC?* *RQ4: How often has LIWC been evaluated in SE research?*, *RQ5: What concerns were raised about adopting LIWC in SE?*

Results: Our findings reveal a wide range of applications, including analyzing team communication to detect developer emotions and personality, developing ML models to predict deleted Stack Overflow posts, and more recently comparing AI-generated and human-written text. LIWC has been primarily used with data from project management platforms (e.g., GitHub) and Q&A forums (e.g., Stack Overflow). Key BSE concepts include *Communication*, *Organizational Climate*, and *Positive Psychology*. 26 of 43 papers did not formally evaluate LIWC. Concerns were raised about some limitations, including difficulty handling SE-specific vocabulary.

Conclusion: We highlight the potential of psycholinguistic tools and their limitations, and present new use cases for advancing research on human factors in SE (e.g., bias in human-LLM conversations).

Keywords: LIWC, Psycholinguistics, Text Analysis, Systematic Mapping Study

1. Introduction

Understanding human factors in SE is critical to improving collaboration, decision making, and productivity in development teams. The vast amount of textual data generated by stakeholders, developers, and users, ranging from code reviews to chats, contains valuable insight into the psychological

and emotional states of those involved in software projects. Understanding these factors can reveal key issues in software teams, such as job satisfaction, project participation, and developer attrition [1, 2, 3]. However, leveraging the full potential of textual data requires the use of specialized tools capable of accurately analyzing unique language patterns and vocabulary used in software engineering [4, 5].

As one of the most widely used psycholinguistic tools, LIWC has played a significant role in advancing research in many scientific fields [6, 7, 8, 9, 10, 11, 12]. In software engineering studies, LIWC has been adopted and used repeatedly by researchers since 2007 until now, providing insight into developer collaboration, the emotional tone of project discussions, the overall dynamics of software teams, etc. While previous studies have examined the use of opinion mining and sentiment analysis tools in SE [13], this study focuses on psycholinguistic tools due to their ability to target and interpret complex psychological constructs. In particular, we explore the significant role of LIWC in the analysis of SE-specific language toward gaining insights into various psychological and social factors affecting the daily tasks of software engineers. Through a systematic paper selection process, we identified 43 papers that study SE-related topics with the help of LIWC. We closely examined each of these papers to answer the following research questions:

- **RQ1: How was LIWC employed in SE studies, and for what specific activities and purposes?**

We found that LIWC has been predominantly used to study *Team Management* in SE, and applied in tasks such as personality detection, emotional tone analysis, and quality assessment of Stack Overflow posts. Moreover,

LIWC has been used both directly and indirectly in these tasks, with the latter including the use of LIWC’s output in feature sets of machine learning models, enabling tasks such as predicting the likelihood of a post’s deletion on Stack Overflow.

- **RQ2: What textual data/datasets were analyzed by LIWC?**

We observed a wide range of textual data types and datasets analyzed using LIWC, such as Q&A posts on Stack Overflow and Kaggle, developer communication on IRC chat channels, and email archives. This diverse set of data sources highlights LIWC’s flexibility in processing different forms of communication in the SE domain.

- **RQ3: What Behavioral Software Engineering (BSE) concepts were studied using LIWC?**

The core BSE concepts studied using LIWC include team dynamics, collaboration, motivation, and developer productivity. Understanding these concepts is critical, as they offer insights into how psycholinguistic tools like LIWC contribute to solving practical SE challenges, particularly by analyzing how communication and collaboration impact team performance.

- **RQ4: How often and in what way(s) has LIWC been evaluated in a software engineering context?**

We identified 17 studies that conducted formal evaluations of LIWC in SE contexts. In our study, evaluation refers to any effort by researchers to assess LIWC’s effectiveness or reliability within their studies. These evaluations varied in scope but provided validation of LIWC’s role in SE research.

We identified 17 studies that conducted evaluations of LIWC in SE contexts. In our study, evaluation refers to any effort by SE researchers to assess LIWC’s effectiveness or reliability. Methods included comparisons with ground-truth data and other forms of methodological validation. These evaluations varied in scope but provided evidence supporting LIWC’s use in SE research.

- **RQ5: What concerns or limitations were mentioned by the researchers adopting LIWC in the software engineering domain?**

We categorized concerns raised about LIWC into two main areas: (1) inherent limitations of LIWC as a lexicon-based tool, e.g., limited language support, and (2) challenges with its adoption in SE, particularly in handling domain-specific vocabulary. These limitations suggest that while LIWC is valuable, there is room for improving its SE-specific vocabulary.

By outlining LIWC’s role in software engineering research, we provide researchers with the insights needed to apply LIWC effectively. This study not only highlights its current applications and limitations, but also encourages the community to enhance and integrate LIWC with other tools for a more holistic understanding of software engineering practices. We discuss opportunities for improving LIWC’s use and propose new approaches, including the integration of LIWC with LLMs, to advance its application in software engineering research.

2. Background

2.1. Psycholinguistics

Psycholinguistics is an interdisciplinary field with roots in psychology and linguistics, investigating the complex connection between the mind and language. More specifically, this discipline seeks to uncover the workings of the human brain when individuals engage in language, be it written or spoken [14, 15, 16].

Psycholinguistics explores a wide range of topics and intersects with several scientific domains. It ventures into the field of neuroscience, studying the physiology of the human brain to understand the functions behind its ability to acquire and use language effectively. Psycholinguists also study the psychological facets of language. Through experimentation, they discover the processes that link language to human cognition, for instance language’s complex interplay with short-term and long-term memory [17].

In addition, the history of psycholinguistics has deep philosophical roots that have sparked transformative moments in the evolution of science. For instance, the development of psycholinguistics contributed to research on language acquisition, which itself has broadened our understanding of the nature of human learning [18]. Similarly, psycholinguistics influenced the shift in the field of psychology from the study of behaviorism to the study of mental processes, for example, language comprehension and generation [19]. A pivotal moment was Noam Chomsky’s critique of behaviorism. The movement he initiated, known as the Chomskyan revolution, significantly affected the field, leaving a mark on the landscape of psycholinguistics [20].

2.2. Text Analysis

According to Neuendorf & Kimberly, *content analysis* can be succinctly defined as “the systematic, objective, quantitative analysis of message characteristics.” [21]. As a specific type of content analysis that focuses on text-based data, text analysis finds application in various domains through tasks including sentiment analysis, topic modeling, information extraction, text summarization, and language translation.

Today, a plethora of text analysis algorithms and software applications are readily accessible. Text analysis algorithms, however, predate the recent breakthroughs in machine learning, deep learning, and natural language processing techniques that have enabled the recent advancements in the quality of text processing. A distinctive characteristic of the prior generation of statistical text analysis algorithms is their utilization of dictionaries, often multiple, created and used for the analysis of text. Dictionaries provide a mapping between specific words in the language and the relevant concepts under study. For instance, the Linguistic Inquiry and Word Count (LIWC) tool leverages custom dictionaries for the measure of the expression of a specific psychological concept in the text, i.e., as a psycholinguistic analysis tool.

2.3. Linguistic Inquiry and Word Count (LIWC)

James W. Pennebaker et al. initially developed Linguistic Inquiry and Word Count (LIWC) ¹ to automate parts of the text analysis process they intended to use for analyzing essays written by people who have experienced

¹<https://www.liwc.app>

trauma. Since then, LIWC has been used across different disciplines to analyze text and, among other use-cases, extract psychological information from the text [22].

In a top-down fashion, LIWC’s dictionaries categorize words into pre-determined categories, mapping them to important psychological constructs and theories. The categories cover a wide range of linguistic and psychological dimensions, including cognition, affect, social processes, culture, and motives, to name a few. [Table 1 lists some of the relevant categories, all of which are still available in the 2022 version of the LIWC.](#)

To improve clarity, we have divided LIWC categories in Table 1 into three subsections as follows: (1) Summary Variables, (2) Text Statistics, and (3) Lexical Variables. The values for *Text Statistics* and *Lexical Variables* are calculated based on the percentage of the words in the text that fall under them. However, *Summary variables* are derived from the percentage of words in other LIWC categories. The Summary Variables stand as the only non-transparent dimensions within the LIWC program, as the precise calculations leading to the values assigned to these categories remain unknown. In Table 1, we include the explanations or common words listed in each category. For example, under *Lexical Variables* section, “Drives” category includes words that are intended to capture motivational language related to affiliation, achievement, and cognitive engagement (e.g., *we*, *know*, *work*).

Based on the word count, i.e., the frequency of specific words in a body of text, and the percentages of words falling into each of the categories, LIWC provides insight into psychological and emotional content in text. Over the years, various studies have validated the correlation between LIWC’s output

and a wide range of psychological concepts. In a 2010 publication, Tausczik & Pennebaker pointed out these correlations by referencing studies that used tools such as surveys, personality assessments, and self-reports to analyze participants' linguistic patterns [22].

Table 1: Examples of LIWC-22 Categories

Category	Description / Frequent Exemplars
Summary Variables	
Analytical thinking	Measure of formal/logical thinking
Clout	Measure of language indicating confidence and status
Authentic	Measure of perceived honesty and genuineness
Emotional tone	Measure of of positive vs. negative affect
Text Statistics	
Word count	Total number of words
Words per sentence	Average words per sentence
Big words	Percentage of words 7 letters or longer
Dictionary words	Percentage of words matched in LIWC dictionary
Total function words	Total number of <i>function words</i> (e.g., <i>the</i> , <i>to</i> , <i>and</i> , <i>I</i>)
Lexical Variables	
1st person singular	<i>I, me, my, myself</i>
1st person plural	<i>we, our, us, let's</i>
2nd person	<i>you, your, u, yourself</i>
3rd person singular	<i>he, she, her, his</i>
3rd person plural	<i>they, their, them, themsel*</i>
Articles	<i>a, an, the, alot</i>
Prepositions	<i>to, of, in, for</i>
Adverbs	<i>so, just, about, there</i>

Category	Description / Frequent Exemplars
Conjunctions	<i>and, but, so, as</i>
Auxiliary verbs	<i>is, was, be, have</i>
Common verbs	<i>is, was, be, have</i>
Impersonal pronouns	<i>that, it, this, what</i>
Numbers	<i>one, two, first, once</i>
Negations	<i>not, no, never, nothing</i>
Positive emotion	<i>good, love, happy, hope</i>
Negative emotion	<i>bad, hate, hurt, tired</i>
Anxiety	<i>worry, fear, afraid, nervous</i>
Anger	<i>hate, mad, angry, frustr*</i>
Sadness	<i>: (, sad, disappoint*, cry</i>
Social processes	<i>you, we, he, she</i>
Work	<i>work, school, working, class</i>
Achievement	<i>work, better, best, working</i>
Leisure	<i>game*, fun, play, party*</i>
Money	<i>business*, pay*, price*, market*</i>
Time	<i>when, now, then, day</i>
Past focus	<i>was, had, were, been</i>
Present focus	<i>is, are, I'm, can</i>
Future focus	<i>will, going to, have to, may</i>
Visual	<i>see, look, eye*, saw</i>
Auditory	<i>sound*, heard, hear, music</i>
Fillers	<i>rr*, wow, sooo*, youknow</i>
Insight	<i>know, how, think, feel</i>
Drives	<i>we, our, work, us</i>
Discrepancy	<i>would, can, want, could</i>
Tentative	<i>if, or, any, something</i>
Certitude	<i>really, actually, of course, real</i>
All-or-none	<i>all, no, never, always</i>

For instance, they highlight how the use of words under the *1st person singular* category e.g., “I”, “me”, and “mine” reflects self-focus. Psychological concepts like self-reflection, insecurity, anxiety, and even depression have been shown to correlate with increased use of *1st person singular* words in various contexts, as individuals experiencing these states tend to center

their attention inwardly, which is reflected in their language use. These findings were established through empirical research that connected participants' word usage with their self-reported psychological states, providing strong evidence of the links between language patterns and psychological phenomena.

Since the initial release of LIWC, multiple versions of it have been created, each characterized by a larger dictionary and a more extensive set of interpretable dimensions. LIWC2015, for instance, uses dictionaries with as many as 4500 words [23], while LIWC-22's internal dictionary contains 12000 words [24]. Notably, new functionalities were added to the 2022 version of LIWC that have been absent in the previous versions. For instance, LIWC-22 offers *narrative arc*, *word frequency and word clouds*, and a *dictionary workbench* designed to streamline the cumbersome process behind the creation of custom dictionaries [24]. Lists of the dimensions and capabilities included in each version of LIWC can be found in the corresponding published journal articles or manuals, e.g., <https://www.liwc.app/help/psychometrics-manuals>.

3. Related Work

3.1. Systematic Literature Reviews of Behavioral Analyses in Software Engineering

Several SLRs were conducted to identify the use of behavioral analyses in SE. The scope of studies incorporating behavioral analysis in SE is extensive, covering topics such as sentiment and emotion detection [13, 25, 26, 27, 28], personality assessment [29, 30, 31], and job satisfaction [3, 32]. Researchers employ a wide range of methodologies, from traditional tools like questionnaires and interviews to automated Machine Learning (ML) models, to an-

answer their research questions. This diversity highlights the vast potential for further exploration to deepen our understanding of the human element in the software engineering process.

Lenberg et al.’s comprehensive literature review on behavioral software engineering, for example, demonstrates the variety of approaches available for studying human factors in SE tasks [33]. Their review points to several key research areas—such as work and organizational psychology, the psychology of programming, and behavioral economics—that contribute to the broader field of behavioral software engineering. This interdisciplinary approach highlights the significance of human behaviors and their influence on software development [33].

Numerous studies have also explored the impact of various human factors within development environments, along with methods for their detection. Lin et al. conducted a systematic literature review (SLR) focusing on opinion mining in software development, including tasks such as sentiment analysis, emotion detection, and politeness detection [13]. Another survey study looked at empirical research on developer emotions between 2005 and 2018 [34]. Felipe et al. conducted a similar review, examining the psychometric tools used to infer personality types in software engineering research [29]. Studies of this nature frequently identify LIWC as a tool for detecting the psychological concepts and human factors they examine. [To date, however, no prior work has systematically reviewed or offered a mapping study on how LIWC itself has been adopted, evaluated, and applied across the entire body of software engineering research. While other secondary studies focus on broader psychological or behavioral techniques, our study is the first to](#)

concentrate exclusively on the role and methodological implications of LIWC within software engineering.

3.2. Use of LIWC Outside Software Engineering

Through a variety of applications, LIWC has demonstrated a unique ability to discover insights about individuals through their written or spoken words. LIWC remains a popular tool in diverse domains of study despite certain limitations inherent to a dictionary-based approach, such as its inability to account for sarcasm or negation in text. [LIWC remains widely used across research domains, with 810 citations in 2023 alone, based on Google Scholar²](#), highlighting its continued relevance and broad impact in various fields.

The work of Pennebaker et al. reflects LIWC’s versatile use-cases [35]. For instance, LIWC has been frequently used to assess the emotional content of text, performing tasks such as sentiment analysis and the detection of specific emotions [36]. Additionally, it has been used to assess the personality traits of individuals who contribute to the text [29]. In psychology research, many have used LIWC to detect depression [6, 37] and to study how the language use of couples relates to their marital satisfaction and the overall health of the marriage [8, 9]. LIWC has also been used to analyze text with political content to assess the public’s opinion on political topics and to find common linguistic patterns in speech of famous politicians [38, 39, 40]. Other researchers have analyzed psychological concepts within a wide range of texts, including suicidal poems [41], Al-Qaeda transcripts [42], and social

²<https://shorturl.at/PPcRR>

media posts written by users dealing with the impacts of the Covid-19 pandemic [43]. The variety and quantity of these studies have showcased LIWC’s unparalleled ability to analyze human emotions, thoughts, and intentions encoded within language.

Despite LIWC’s wide adoption across many disciplines, and its frequent appearance in SE studies, no existing study has examined its specific applications, limitations, and potential, specifically within the context of software engineering. This gap motivates the need for a focused SLR that examines LIWC’s contributions to SE research, an effort we undertake in this study.

4. Methodology

The goal of this paper is to study and analyze the software engineering research that uses LIWC. Following the methodology guidelines offered by Kitchenham et al. [44], this section outlines our approach toward identifying relevant research papers. An overview of our approach for selecting the relevant papers is shown in Figure 1. Additionally, we address all PRISMA items in Kitchenham et al.’s checklist for secondary studies to ensure alignment with the standards [45]. For further reference, please find our PRISMA 2020 checklist attached in the Appendix section.

To select the papers for our survey, we first curated a query to retrieve as many relevant research papers in digital libraries as possible. We examined the retrieved papers closely (first title and abstract and then the full text) to filter out any irrelevant or inappropriate studies based on a set of specific inclusion and exclusion criteria (detailed description in Table 3). This step was followed by forward and backward snowballing in order to locate any papers

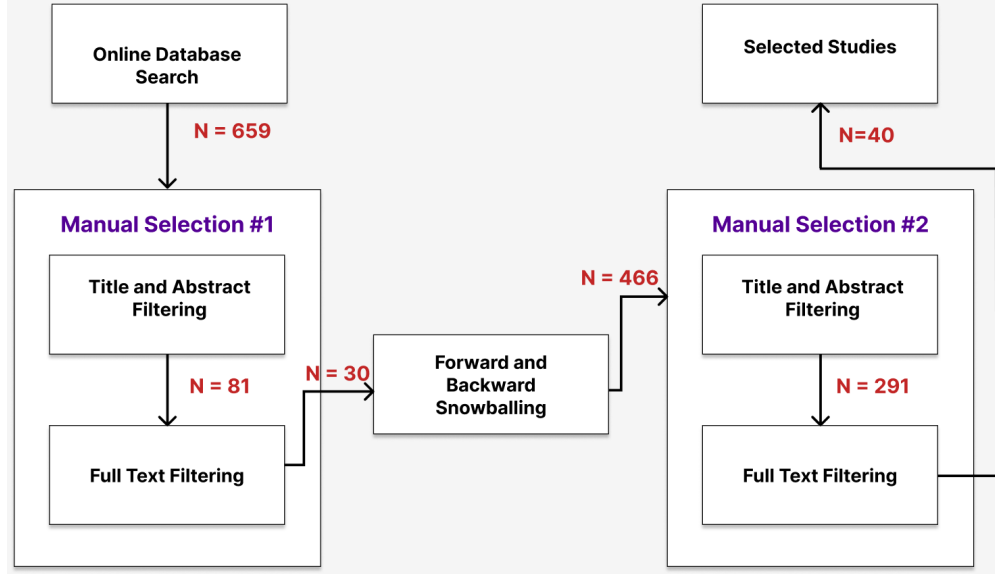


Figure 1: The process of relevant study identification.

that our initial query may have missed. Specifically, we adopted a hybrid approach that combined comprehensive keyword-based searches with both forward and backward snowballing, as supported by recent SLR methodologies [46, 13, 34]. Finally, the newly collected papers were again manually filtered, leaving us with the final set of papers that we analyzed in this study ($n=43$). The remainder of this section explains each step of this process in more detail.

4.1. Search Strategy & Online Query

To ensure a comprehensive retrieval of relevant studies, we refined our search query iteratively through a process of trial and validation. Initially, we started with broad queries such as “liwc” OR “linguistic inquiry and word

count”, which retrieved a large number of papers across multiple digital libraries, but many were unrelated to software engineering. We then progressively refined the query by adding domain-specific terms, such as “software”, “software engineering”, and “developer” OR “development”, to filter studies focused on software engineering applications. For instance, adding “software engineering” significantly reduced irrelevant results while maintaining coverage of key studies. Additionally, we explored broader psycholinguistic and behavioral software engineering terms, such as “psycholinguistic” OR “psychometric” AND “text analysis” AND “software engineering”, to identify studies that might not explicitly reference LIWC but were relevant to our research scope. The final query (“liwc” OR “linguistic inquiry and word count”) AND (“software engineering”) retrieved the highest number of relevant papers possible. A full record of all queries, retrieval counts, and query refinements is included in our supplementary material for transparency and reproducibility.

This process, coupled with insights from prior literature regarding suitable phrases for the retrieval of the software engineering papers [13], led us to select the following search query:

(“liwc” OR “linguistic inquiry and word count”) AND (“software engineering”)

The initial part of the query, (*“liwc” OR “linguistic inquiry and word count”*), is designed with the intent of inclusively capturing papers that reference the complete name of the targeted tool or its widely recognized abbreviation, “LIWC”. The latter component of our query, (*“software engi-*

neering”), serves as a filter, enabling the identification of papers discussing subjects related to the field of software engineering while not posing as much limitations as terms such as “developer”.

We selected prominent digital archives for our study: IEEE Xplore Digital Library [47], ACM Digital Library [48], Elsevier ScienceDirect [49], Springer Link Online Library [50], Wiley Online Library [51], and Scopus [52]. The questionable quality of some content on Google Scholar led us to the decision of excluding it as a resource [53]. [The aforementioned search string was used in all online libraries without modifications. Table 2 illustrates the number of papers retrieved from each source in our study in August of 2024.](#)

Table 2: Number of Retrieved Papers From Each Source

Source	Number of Retrieved Papers
IEEE Xplore Digital Library	6
ACM Digital Library	88
Springer Link Online Library	444
Elsevier ScienceDirect	56
Wiley Online Library	8
Scopus	279
Total (after duplicate removal)	788

After eliminating duplicate papers from various sources, we were left with a total of 788 unique documents. Given both the reasonable number of papers and our resource constraints, we decided not to impose any further restrictions on the searches carried out within these online libraries. Our approach involved conducting the searches through the entire body of text rather than limiting ourselves to titles and abstracts alone. Although this method may have led to a higher number of unrelated papers in our results, we deemed it essential to ensure that we identified all relevant papers. This

choice was made because not all studies that utilize LIWC at some point in their process explicitly mention it in their titles or abstracts. Next, all 659 articles were subjected to a thorough manual examination, following the method detailed in Section 4.2, to pinpoint the relevant studies for this research.

4.2. Paper Eligibility and Selection

4.2.1. Inclusion & Exclusion Criteria

To support the manual paper selection process, we first established a precise set of inclusion and exclusion criteria in Table 3, following the PRISMA framework’s requirements [45]. All the papers included in this study must have undergone peer-review and have been published in conferences, workshops, or journals. These papers primarily focus on topics related to software development and use LIWC in their analyses.

4.2.2. Manual Paper Selection

Two authors contributed to this manual selection process. The first author used the inclusion and exclusion criteria to identify relevant papers. After each of the two review phases, the second author independently examined the final list, verifying whether each paper’s inclusion aligned with the criteria. Discrepancies or uncertainties were discussed between the two authors until full agreement was reached. *This process was repeated after the initial search as well as the snowballing process, adding a layer of rigor and helping ensure a consistent application of our selection criteria.*

Our manual selection process began by inspecting the titles and abstracts

Table 3: Inclusion and Exclusion Criteria

Inclusion Criteria	
IC1	The paper must have undergone the peer-review process and published at conferences, workshops, or journals
IC2	The paper must be available in one of the online libraries mentioned in Section 4
IC3	The conducted study in the paper must contribute to our understanding of the software development process or propose a solution to further advance the related practices.
IC4	The study must adopt LIWC to analyze textual data.
Exclusion Criteria	
EC1	The paper is not entirely written in English, the dominantly used language for research publications
EC2	The adopted lexicons and methods in the research are specific to a language other than English, which makes the technique not comparable to others.
EC3	The paper either serves as an extension of a previously accounted-for conference publication or its extended version has already been included.
EC4	The paper is not a traditional research publication and instead serves as an abstract-only submission, doctoral thesis, book, poster, etc.
EC5	The paper does not explain what approach was used and in what way LIWC was used.
EC6	Papers that do not meet the inclusion criteria of contributing to SE and using LIWC.

of the papers in order to eliminate irrelevant ones, such as those unrelated to software engineering research, guided by our exclusion criteria. In cases where the information in the title and abstract did not provide sufficient grounds for exclusion, the paper was kept to undergo the next step of our manual selection.

Close examination of titles and abstracts resulted in a collection of 91 potential papers. Subsequently, we examined the text of these 91 papers in their entirety, applying both inclusion and exclusion criteria to determine their eligibility for inclusion in our study. At this stage, we also took note of any information necessary to answer the research questions, e.g., the data types, the data sets and the LIWC categories used in the study. The full-text assessment led to the retention of 36 relevant studies.

4.2.3. Snowballing Process

As a measure to mitigate any potential limitations of our initial online digital library search, we adopted a forward and backward snowballing approach on the 36 selected papers. Snowballing was performed by the first author and involved the gathering of all the citations and references cited within these 36 papers, thereby ensuring the inclusion of additional potential papers that may not have been found in our initial search results.

Our snowballing process led to the identification of 494 new and unique papers, which we then subjected to manual selection. Based on the titles and abstracts, we narrowed the number down to 301 potential papers. After conducting a thorough full-text-based filtering by the first author and verification by the third author, we finally identified 7 additional papers.

4.3. Data Extraction and Synthesis

In this section, we describe the approach used to extract and code data from each paper included in our study. After retrieving the papers from the sources discussed above, we automatically collected basic metadata for each paper, including its title, author(s), publication year, and venue i.e., journal or conference. This information, along with the papers themselves, is made available in our supplementary material ³.

In line with SLR requirements and in order to systematically analyze the content of each paper, we created a data extraction form relevant to our research questions. We used this form to record data such as the LIWC categories utilized, the datasets analyzed, the specific software engineering activities addressed and the behavioral software engineering concepts investigated. The list of all data fields required is presented below:

1. Paper title
2. Authors
3. Year of publication
4. Publication venue (journal/conference)
5. Purpose for using LIWC
6. LIWC utilization: direct interpretation or indirect use
7. LIWC categories used
8. Dataset(s) analyzed
9. Software engineering activities studied

³https://figshare.com/articles/dataset/Psycholinguistic_Analyses_in_Software_Engineering_Text_A_Systematic_Literature_Review/28774319

10. Behavioral software engineering concepts addressed
11. Evaluation methods used to assess LIWC's effectiveness
12. Mentioned limitations or concerns with using LIWC

For identifying the software engineering activities and behavioral software engineering (BSE) concepts, we adopted a deductive coding approach [54], applying established taxonomies as discussed in Sections 5.3 and 5.4. For other data extraction tasks, such as identifying the purpose of using LIWC, types of data analyzed, and concerns reported by authors, we employed an inductive coding approach [54], developing categories iteratively based on recurring patterns in the reviewed papers. All data extraction tasks, including both deductive coding (e.g., software engineering activities, BSE concepts) and inductive coding (e.g., evaluation methods, limitations, LIWC usage purposes), were performed independently by two authors to ensure consistency. For fields requiring synthesis without predefined categories, specifically evaluation methods, limitations, and purposes of LIWC usage, the first author conducted thematic analysis to identify patterns and categories, after which the results were reviewed and verified by the second author. Two authors independently coded all 43 studies. Following Kitchenham et al.'s guidelines for secondary studies [44], we calculated Cohen's Kappa coefficients for each deductive coding dimension to assess inter-rater reliability. The resulting Cohen's Kappa coefficients were: 0.61 for both software engineering activities and BSE concepts, and 1.00 for determining the direct/indirect use-cases of LIWC, the purpose for using LIWC, datasets used, evaluation techniques, and concerns or limitations. Where initial agreement was below the threshold

of substantial agreement (i.e., less than 0.6), the coding was refined through discussion until substantial agreement was achieved across all dimensions, following the guidelines proposed by Landis and Koch [55]. The slightly lower agreement for SE activities and BSE concepts can be attributed to the inherent complexity and occasional overlap of categories and labels within each of these dimensions’ taxonomies, which may lead to subtle differences in interpretation. Nonetheless, the final levels of agreement, each above 0.6, meet the standard for substantial inter-rater reliability.

5. Results

5.1. *Final Set of Papers*

In total, our paper selection process led to the identification of 43 studies (36 in the first iteration + 7 through snowballing) that are the subjects of this survey. The complete set of final papers used throughout this work is listed in Appendix A, where references to each study are provided. Henceforth, we refer to these papers with “S” prefixes e.g., [S1], [S2], etc. Table 4 illustrates the distribution of our chosen papers based on their publication years. The first software engineering paper using LIWC was published in 2007 [S27]. Since then several software engineering studies have used LIWC to directly analyze many facets of developers’ communications. After 2013, the number of software engineering publications that leveraged LIWC in their studies has noticeably grown. As of 2023, LIWC continues to feature prominently in software engineering studies, highlighting the community’s ongoing interest and the relevance of this tool in the research area.

Table 4: The Use of LIWC in Software Engineering Research Over Time

Year	Number of Papers	Running Total
2007	1	1
2012	3	4
2013	5	9
2014	5	14
2016	5	18
2017	4	23
2018	3	26
2019	1	27
2020	4	31
2021	4	35
2022	4	39
2023	3	42
Jan-Aug 2024	1	43
Total	43	43

5.2. *RQ1: How was LIWC employed in SE studies, and for what specific activities and purposes?*

5.2.1. *Software Engineering Activities*

We examine specific software engineering activities that were studied using LIWC in SE studies. Borrowing Lin et al.’s taxonomy [13], we categorized the studies based on the following development activities: *Design Definition Process*, *Knowledge Management Process*, *Quality Assurance Process*, *Stakeholder Needs and Requirements Definition Process*, *Team Management*. Most papers tackled subjects related to the *Team Management* in development teams. Despite the heavy focus on the human interactions of developers and team members, the studies had different focus areas. For instance, three of the papers that used LIWC to process app reviews focus on *Quality Assurance* activities [S22, S38, S40] and three other studies focus on the *Knowledge*

Management Process [S1, S15, S39]. Furthermore, Hellman et al.’s work on analyzing the language used in user forums of projects such as Zotero and Audacity examines the behaviors and expectations of the end-user and therefore makes a contribution to the *Stakeholder Needs and Requirements Definition Process* [S21].

It does not come as a surprise that LIWC, a tool frequently used for automated psycholinguistic analysis, is being used dominantly to better manage developer teams. The more creative approaches can link LIWC’s results to the more technical aspect of software development; however, tangible improvements to the software engineering process are not only possible through technical advances [56]. Additionally, many studies point to the existence of correlations between team members’ behaviors and projects’ outcomes which can be essential to the improvement of the software engineering processes [57, 58, 59].

5.2.2. *Direct and Indirect Use of LIWC*

As part of answering RQ1, we examined how LIWC has been used across SE studies and identified two overall usage patterns: *direct* and *indirect*. In Table 5, we list all the direct and indirect uses of LIWC along with the categories used for each performed activity and task. We elaborate on both usage types below.

Among all the included studies in our survey, 19 used LIWC to directly analyze text. As the name suggests, direct use of LIWC refers to the cases in which researchers used LIWC’s outputs directly to interpret and gain insights from the textual data without any further processing. These use-cases frequently encompass tasks such as detecting developer emotions (e.g., anger,

negative emotion, and anxiety) [S13, S18, S28, S30] using LIWC categories like *affect*, *emotions*, etc. Other tasks included assessing characteristics, such as goal orientation and sociability [S5, S10, S23, S26] and evaluating the linguistic features of LLM responses to Stack Overflow questions.

On the other hand, 21 studies employed LIWC’s output indirectly, using it to develop other algorithms or models, or to extract insights that necessitated further processing of LIWC’s outputs. The most common method of using LIWC indirectly was to uncover developers’ personalities [S2, S8, S9, S19, S27, S35]. Indeed, prior research has demonstrated multiple methods of inferring text authors’ personalities using LIWC. Pennebaker & King [60], Yarkoni et al. [30], and Goldbeck et al. [61] all have proposed methods of leveraging LIWC for measuring the personality traits of an author based on the widely used Big-Five Factor personality model [62]. However, it should be noted that while the original studies that employ LIWC categories to measure personality traits have received significant attention [60, 30, 61], their focus has predominantly centered on general rather than domain-specific text, such as software engineering-related content.

Table 5: The Direct (D) and Indirect (I) Use of LIWC and The Specific Categories Used for Text Analysis.

Activity	Task	D/I	LIWC Categories	Study
Team Management	Group Communication Analysis in Teams	D	Pronouns, Cognitive, Work and Achievement, Leisure, Social and Positive, Negative Language	S26, S31, S32, S34, S36
	Group Communication Analysis in Teams	D	Cognitive, Work and Achievement, Leisure, Social and Positive, Negative Language	S29, S33
	Identifying Developer Preferences	I	Visual, Auditive, “Kinaesthetic”	S25

Continued on the next page

Activity	Task	D/I	LIWC Categories	Study
	Group Communication Analysis in Teams	D	Achieve, Affect, Discrep, Funct, Hear, Humans, Leisure, Negative Emotion, Numerals, Past, Present, QuestionMark, Social	S18
	Group Communication Analysis in Teams	D	Work and achievement, negative, cognitive, social and positive	S17
	Group Communication Analysis in Teams	D	Linguistic process, Linguistic categories, Psychological Processes, relativity, current concerns, spoken categories, and punctuation	S6
	Group Communication Analysis in Teams	D	Pronouns, Cognitive, Work and Achievement, Emotions	S5
	Group Communication Analysis in Teams	D	Pronouns, Work and Achievement, Cognitive, Leisure, Social and Emotions	S10
	Group Communication Analysis in Teams	D	Achievement, Work, Positive, Negative, Insight, Social, prop(I)	S37
	Group Communication Analysis in Teams	D	prop(focusfuture), prop(we), prop(posemo), prop(negemo), prop(certain), prop(tentative)	S9
	Group Communication Analysis in Teams	I	<i>Categories Not Specified</i>	S4
	Group Communication Analysis in Teams	I	Personal pronouns, Impersonal Pronouns, Articles, Prepositions, Auxiliary verbs, Common adverbs, Conjunctions, Negations, Analytical Thinking, Clout, Authentic, and Emotional Tone.	S24
	Group Communication Analysis in Teams	D & I	Standard WC Pronouns, Cognitive, Relative, Optimism, References to Time, Future tense verbs, Social Processes, Categories correlated with personality traits in [60]	S27
	Personality Detection	I	Categories correlated with personality traits in [60]	S2, S20, S23
	Personality Detection	I	Categories correlated with personality traits in [30] and [61]	S19
	Personality Detection	I	Categories correlated with personality traits in [30]	S7
	Personality Detection	I	Categories correlated with personality traits in [30], Categories used in Personality Recognizer tool ⁴	S8
	Personality Detection	D & I	Categories correlated with personality traits in [60], Cognitive, Work and Achievement, Leisure, Social and Positive, Negative Language	S35
	Emotion Detection	D	Positive affect (Positive Emotion), Negative affect (Negative Emotion, Sadness, Anger, Anxiety, Inhibition)	S30
	Emotion Detection	D	Anger	S28

Continued on the next page

⁴<http://s3.amazonaws.com/mairesse/research/personality/recognizer.html>

Activity	Task	D/I	LIWC Categories	Study
	Emotion Detection	I	<i>Categories Not Specified</i>	S12, S13
	Emotion Detection	I	Anger	S14
	Sentiment Analysis	I	<i>Categories Not Specified</i>	S3
	Argument Extraction	D & I	<i>Categories Not Specified</i>	S11
Quality Assurance	Predicting Deletion of Stack Overflow Post	I	Personal Pronouns, Pronouns, Space, Relativity, Inclusive, Cognitive Process, Social, 1st Person Singular Pronouns, Function Words, Conjunctions, Prepositions	S1, S39
	LLM Evaluation	D	Affective Process, Cognitive Process, Perceptual Process, Informal language	S43
	Insider Threat Detection	I	<i>Categories Not Specified</i>	S41
Stakeholder Needs and Requirements Definition	User Input Analysis	D	Clout, Authentic, Analytic, Tone, Cognitive process, Affective process, Drives	S21
	User Input Analysis	I	Negative Emotion	S15, S22, S38, S40
Design Definition Process	Design Bias Analysis	D	Personal Concerns (“persconc”), Social and Affect	S42

The entries of Table 1 provide further insight into all the categories mentioned in our selected papers.

5.2.3. Notable Applications of LIWC

Through answering RQ1, we also aim to better understand the types of LIWC applications within the SE research. We pinned down the purposes LIWC has fulfilled in each study. Here, in more detail, we discuss some notable applications of LIWC in SE research.

To predict the likelihood of a post’s deletion on Stack Overflow, researchers have used LIWC to analyze its content. Given that many Stack

Overflow posts can be either off-topic or low quality (predominant reasons of deletion [63]), the task of manually sifting through the posts and deleting such content is laborious and time-consuming. Consequently, the development of automatic methods for identifying posts that need to be deleted can offer significant support to those responsible for monitoring the website’s content. Correa & Sureka created a classification model trained on 47 features, some of which were created using the textual contents of the posts. Within these features were 11 LIWC categories such as *Relativity*, *Cognitive*, *Inclusive*, and *Social* that were considered potentially distinctive for this task. Using these features, a prediction model with 66% accuracy was developed. Additionally, this study found new insights into the quality of the user-generated content and characteristics of the deleted posts on Stack Overflow. This study was later built upon and improved in another work where the 47 features were reused in a different model to improve the prediction of Stack Overflow posts that need to be deleted [S39].

LIWC has also been used in the development of other psychometric tools. In another work, Santos et al. present *NEUROMINER*, a psychometric tool based on LIWC, that identifies developers’ preferred representational systems [S25]. Representational systems refer to the ways in which the human brain stores and processes information. Often within the NLP domain, these systems are recognized as *Visual*, *Auditive*, or *Kinaesthetic* [64]. Knowing the developers’ preferred representational systems can help us better understand the most effective methods of communication and knowledge sharing. *NEUROMINER* extracts LIWC values from an input text, and performs ANOVA (analysis of variance) to detect the preferred representational systems of the

top four Apache committers. The authors use *Visual*, *Auditive*, and *Kinaesthetic* categories of LIWC to decide which of these representational models a developer tends to prefer.

Zolduoarrati et al. studied the impact of cultural differences in developer communications with the help of LIWC. The authors conducted a study, examining Stack Overflow users' behaviors across three distinct countries, characterized by varying degrees of individualistic or collectivist values [S10]. The culture of the selected countries i.e., the United States, China, and Russia, exhibit different levels of individualism, as determined by Hofstede's cultural index [65]. The authors argue that understanding the cultural patterns of users from these countries holds potential for improving communication and cooperation in globally distributed software development efforts. To accomplish this, the study leveraged LIWC categories, including *Cognitive*, *Emotional*, and *Leisure* dimensions, to analyze the text written by the users from these countries.

Furthermore, researchers have used LIWC to examine the topic of trust among developers in a collaborative team. Even though some studies on this topic exist [26, 66, 67], few attempted to automatically measure trust [68, 69]. Wang & Redmiles have, however, combined NRC Lexicon [70] and some dimensions of LIWC related to trust to measure a baseline of trust among developers [S4]. They did not specify the exact LIWC categories used; nevertheless, they evaluated their findings by calculating the baseline level of trust with NLP methods based on the communications of 10 individuals involved in their studied projects.

Janjua et al.'s study addresses the growing challenge of insider threats in

information technology systems, where insiders can bypass security measures and access confidential documents [S41]. In their work, they utilized the content of email communications and the output from LIWC analysis to train various machine learning models aimed at detecting insider threats.

Also worth mentioning is Han et al.’s study on the cross-status communications of the developers in open source software development [S24]. In their work, the authors use 12 LIWC categories to compute the similarities of the language styles used by different groups of developers and explore the possibility of a correlation between the project outcomes and the cross-status communications between *elite* and the *non-elite* developers.

Additionally, [S19] and [S8] examine LIWC’s ability to infer developers’ personalities and compare the tool with the now-discontinued LIWC-based tool, IBM Personality Insights ⁵. While [S19] finds the two tools to produce consistent results, [S8] points out a noticeable level of disagreement between LIWC and IBM Personality Insights.

Finally, in attempting to analyze the opinions of the community in open source software development, LIWC has been utilized to create the Argulens framework [S11]. Argulens creators explore how usability discussions occur in issue tracking systems, and how effective the machine learning models can be in extracting arguments. They also discuss the potential of the framework to support practitioners in understanding community opinions. LIWC’s output is used alongside other features as an input to the machine learning classifiers for tasks such as classifying the Argumentative vs. Non-argumentative

⁵<https://watson-developer-cloud.github.io/node-sdk/master/classes/personalityinsightsv3.html>

conversations.

RQ1 Summary: LIWC has been directly and indirectly utilized in a variety of SE tasks to understand the nuances of developer learning, collaboration, and communication. The most common uses include personality detection and emotion analysis. Other applications include quality assessment of forum posts, investigation of cultural differences in distributed teams, and argument extraction.

5.3. RQ2: What types of textual data were analyzed using LIWC in SE studies?

To categorize the software engineering texts analyzed with LIWC, we identified the types of developer communication used in each study. Table 6 shows the data types and sources of datasets used in the 43 papers we examined.

Most researchers focused on analyzing the communications of developers on project management platforms, i.e., IBM Rational Jazz, GitHub, and Jira. In 10 of the 43 studies, IBM Rational Jazz was used as the primary source of data. In addition to hosting developers' communications, IBM Rational Jazz provides environments for both development and tracking the progress of the project. These 10 papers use LIWC to study developers' comments in Jazz repositories over time, and to explore various aspects of developer interactions and social dynamics, including the impact of leadership and knowledge sharing on project success [S26, S31]. Figure 2 provides a visual overview of the distribution of data sources across the studies we reviewed.

Q&As are another type of commonly used data that is analyzed with the help of LIWC. Of the 11 studies that have focused on Q&A forums [S6, S21],

Table 6: Data Types and Sources

Data Type	Sources	Studies
Interactions on project management platforms	IBM rational jazz	S2, S17, S26, S29, S31, S32, S33, S34, S36, S37
	GitHub issues	S4, S7, S9, S11, S14, S19, S24, S28
	GitHub pull requests	S7, S9, S19, S24, S28
	GitHub code reviews	S7, S24, , S9, S19
	Jira issues	S12, S13
	GitHub discussions	S4
Q&A	Stack Overflow	S1, S3, S5, S10, S20, S23, S35, S39
	Kaggle forums	S6
	ChatGPT answers to Stack Overflow questions	S43
	Specific software product forums (VLC, Zotero, etc.)	S21
Email Communications	Apache Software Foundation mailing lists	S4, S8, S16, S25, S27
	Malicious users' interactions	S41
User Feedback	App Reviews	S15, S22, S38, S40
Chats	IRC	S4
Microblogs	Controlled Studies	S18
	OfficeTalk software (developed exclusively for internal use in Microsoft)	S30
Written Solutions to Design Tasks	Conducted Experiments	S42

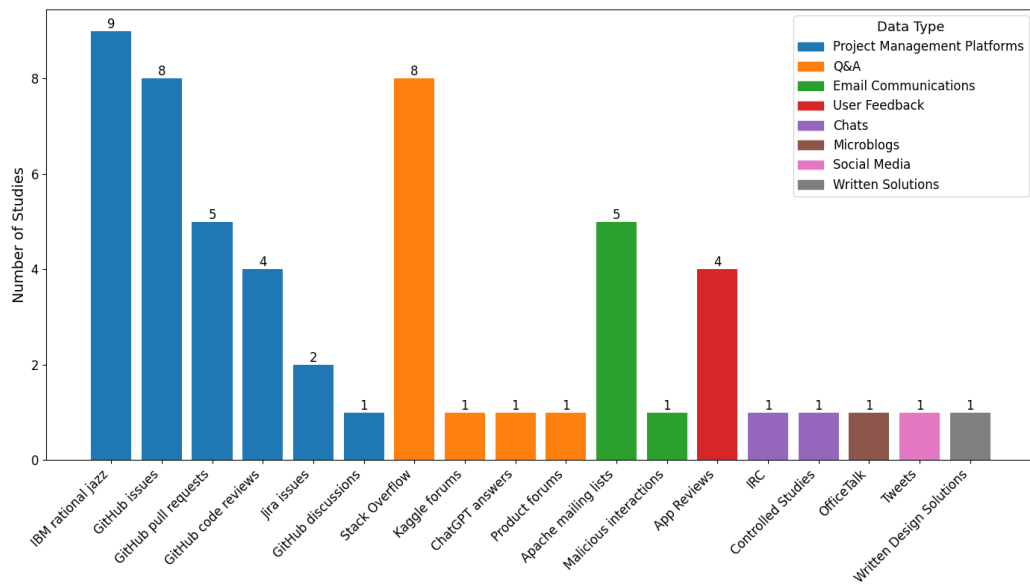


Figure 2: Distribution of Data Sources Used in SE Studies Employing LIWC. The figure highlights the prevalence of project management platforms and Q&A forums as dominant sources of analyzed text.

the most popular Q&A platform, Stack Overflow, has been used in 8 different studies. While some studies using Stack Overflow data focus on personality detection [S20, S23, S35], other studies use the data to identify low quality posts [S1, S39], different attitudes of developers based on their cultural background [S10], developer sentiment [S3], and in one case, exploring differences in the attitudes of male and female developers [S5]. Additionally, one study used Stack Overflow questions as prompt for ChatGPT, in order to compare the LLM-generated answers to those provided by users [S43].

The Apache Software Foundation (ASF) mailing lists have also served as data sources for a number of studies. ASF mandates that projects must maintain full transparency in their communications through public mailing lists, following the commonly held principle that, if it wasn't documented on the mailing lists, it didn't happen [71, 72]. By requiring all the developer communications to take place on mailing lists, ASF creates a valuable source of data for those interested in analyzing the interactions of many ASF projects' team members. Such analyses were used to study the trust dynamics among the developers, developers' behaviors, and developers' personalities [S4, S8, S27].

Four studies used LIWC to analyze app reviews [S15, S22, S38, S40]. All 4 of these studies used LIWC specifically to identify the words associated with the negative emotions according to LIWC's lexicon. The findings were then combined with other observations and data points extracted from the reviews. Authors of these 4 papers use the findings to suggest prioritizing the improvement of certain app features over others, draw conclusions about user satisfaction with an app, etc.

Notably, a few studies used LIWC to analyze software engineering chats and microblogs [S4, S18, S30]. Given the prevalence of developers’ emotional expressions in chats [27, 4, 73], exploring various emotional and psychological factors through developers’ chat communications may yield novel insights into the human factors affecting the development process.

RQ2 Summary: The most frequent data sources include project management platforms (e.g., GitHub) and Q&A forums (e.g., Stack Overflow). Other data sources include mailing lists, app reviews, and developer chats. This diverse set of data sources highlights LIWC’s flexibility in processing different forms of communication in the SE domain.

5.4. RQ3: What Behavioral Software Engineering (BSE) concepts were studied using LIWC?

Behavioral Software Engineering (BSE) is defined as “the study of cognitive, behavioral and social aspects of software engineering performed by individuals, groups or organizations.”[33]. BSE combines social sciences and empirical software engineering to gain insight into human factors in software engineering [29, 33, 74]. Through their work, including studying psychology textbooks, examining related papers, and interviews with field experts, Lenberg et al. identified 55 core concepts underpinning BSE research, such as *Group Norms*, *Organizational Climate*, and *Leadership* [33].

Understanding how LIWC has been used to study BSE concepts reveals the tool’s effectiveness in capturing critical human factors that directly influence software engineering practices and outcomes. Answering this research question contributes to assessing LIWC’s role in advancing behavioral re-

search within SE. Next we discuss in detail the studied behavioral software engineering concepts using LIWC in our 43 selected studies.

To find out whether or not LIWC has been used to conduct software engineering-related studies tackling different BSE concepts, we have identified all the BSE concepts studied in each of our selected studies. Table 7 illustrates the definitions of the BSE concepts that have been borrowed from [33] and identified in one or more of our selected 43 studies.

Table 7: BSE Concepts found in our study

BSE Concept	Definition	Studies
Communication	“The process that allows people to exchange information, feeling or thoughts ” [33, 75]	All selected studies
Organizational Climate	“Is defined as the recurring patterns of behavior, attitudes and feelings that characterize life in the organization (Isaksen et al., 2007; Denison, 1996).” [33, 76, 77]	S4, S5, S7, S12, S14, S24, S27, S28, S29, S30, S31, S32, S33, S34, S36, S37
Motivation	“According to Pardee (Pardee, 1990) motivation can be defined as those forces within an individual that push or propel him to satisfy basic needs. It is what prompts a person to act in a certain way or at least develop an inclination for specific behavior.” [33, 32]	S7, S9, S12, S17, S21, S22, S26, S31, S33, S36, S37, S38, S40
Positive psychology	“Positive psychology is the branch of psychology that study the strengths and virtues that enable individuals and communities to thrive (Seligman and Csikszentmihalyi, 2000).” [33, 78]	[S3, S17, S18, S23, S26, S27, S28, S29, S30, S31, S33, S35, S37
Personality	“According to Feist and Feist (Feist, 1994) personality is a pattern of relatively permanent traits and unique characteristics that give both consistency and individuality to a person’s behavior.” [33, 79]	S2, S7, S8, S17, S19, S20, S27, S35, S37

Continued on next page

Table 7 – Continued from previous page

BSE Concept	Definition	Studies
Leadership	“Is the art of influencing followers to achieve success by identifying joint goals, finding best-fit roles in teams, collaborating constructively and dynamically, and adapting to change within their environments (Wikipedia, 2014).” [33, 80]	S9, S18, S27, S29, S24, S31
Group dynamics	“Is a system of behaviors and psychological processes that occurs within a group or between groups (Forsyth, 2009).” [33, 81]	S2, S4, S6, S10, S11, S24, S28
Group norms	“Are informal rules that regulates the behavior of the group (Feldman, 1984).” [33, 82]	S5, S6, S24, S36
Cognitive style	“Individual differences regarding strategies for perceiving, remembering, thinking, and problem solving” [33, 83]	S16, S25
Intentions to leave	“Refers to conscious and deliberate willfulness to leave the organization (Tett and Meyer, 1993).” [33, 84]	S27
Organizational Change	“Is both the process in which an organization changes and the effects of these changes on the organization (Portal, 2014). ” [33, 85]	S27
Group social identity	“Defined by Hogg and Vaughan as an self-concept derived from perceived membership of social groups, i.e., the aspects of a person that are defined in terms of his or her group memberships (Hogg and Vaughan, 2002).” [33, 86]	S10
Risk taking	“Is the tendency to engage in potential harmful behavior that, at the same time, could have some kind of positive outcome (March and Shapira, 1987).” [33, 87]	S41
Stress	“Defined by the American Psychological Association as a transient state of arousal with typically clear onset and offset patterns” [33, 88]	S14

Communication, in particular stands out as the single most studied BSE concept as the researchers employing LIWC consistently analyze various forms of textual communication, whether among developers themselves or between other stakeholders and developers, as a means of discovering new information.

Organizational Climate, Motivation, Positive Psychology, and Personality represent the most extensively explored concepts within the collected body

of literature. Among our subjects of study, the least commonly studied BSE concepts were *Risk taking*, *Stress*, and *Group Dynamics*.

In line with Lenberg et al.'s results, indicating *Communication* and *Personality* to be among the most studied BSE concepts, we found both of these concepts to be frequently examined in papers employing LIWC in software engineering. However, contrary to the observations by Lenberg et al., we found that *Organizational Climate* and *Positive Psychology* also to be frequently examined and studied in a number of papers. Figure 3 illustrates the most commonly studied BSE concepts in our selected literature.

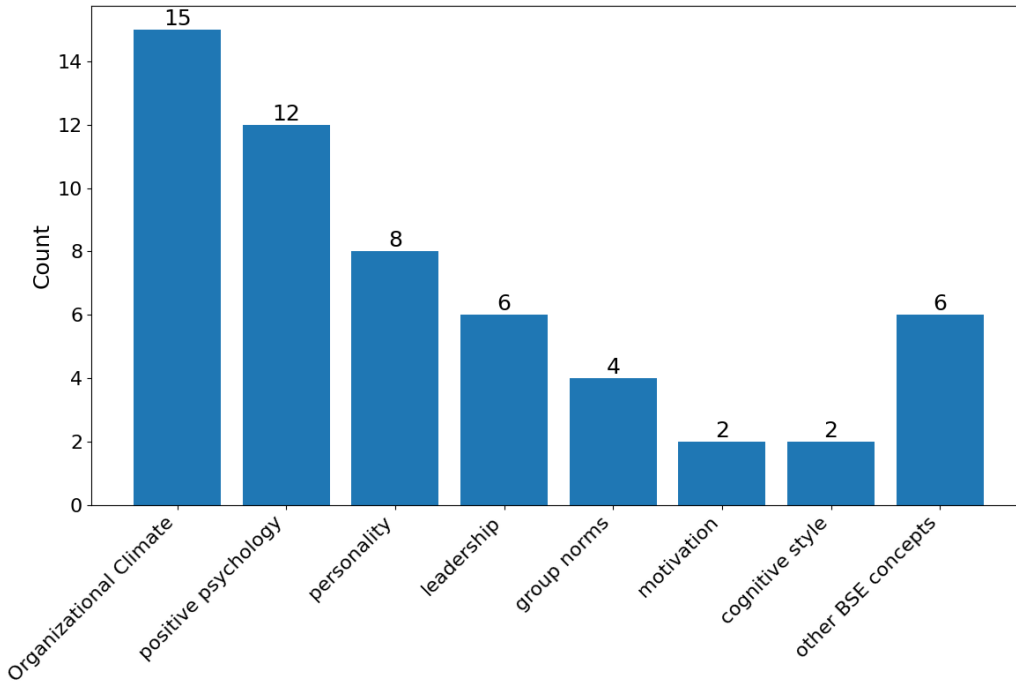


Figure 3: Most Common BSE Concepts Studied in Software Engineering Literature Using LIWC.

While some BSE concepts, such as sentiment or emotion, have long been

studied using a range of automated tools, others like *Organizational Climate* have traditionally relied on qualitative methods, such as interviews or surveys, due to their abstract and contextual nature. However, LIWC has enabled researchers to examine these social concepts in a more scalable and automated fashion through text analysis. This may help explain why *Organizational Climate* appears more frequently in our set of studies than reported in earlier BSE reviews [89]. Similarly, BSE and social concepts like *Stress* and *trust*, which might otherwise be difficult to quantify or observe without self-reporting, become more accessible through LIWC’s built-in stress-related categories. In this way, LIWC lowers the barrier to analyzing many complex psychological or social phenomena in SE, helping to expand the reach of BSE research.

RQ3 Summary: We present all the BSE concepts studied with the use of LIWC in Table 7. The most commonly examined concepts include *Communication*, *Motivation*, *Organizational Climate*, *Positive Psychology*, and *Personality*, while concepts such as *Risk Taking*, *Stress*, and *Group Dynamics*, are less commonly studied.

5.5. RQ4: How often and in what way has LIWC been evaluated in a software engineering context?

Like any automated analysis tool, LIWC must be evaluated in order for us to determine its effectiveness within the context of SE studies. In the context of our study, evaluation refers to any effort by SE researchers to assess the effectiveness or reliability of LIWC in their studies. Such evaluations could involve expert assessments, for example, a psychologist reviewing whether LIWC is accurately capturing relevant psycholinguistic signals, or

Table 8: List of Papers With and Without Evaluation methods for LIWC’s Output

Evaluation Method	Papers	% of Studies
No Evaluation	S2, S5, S6, S7, S10, S11, S16, S17, S18, S20, S21, S22, S23, S24, S26, S28, S31, S32, S33, S34, S35, S36, S40, S41, S42, S43	60.5%
Indirect Evaluation	S1, S3, S4, S8, S12, S13, S14, S15, S19, S25, S27, S29, S30, S37, S38, S39	37.2%
Comparison with Other Models	S9	2.3%

comparisons between LIWC output and established ground truth data. To answer RQ4, we identify and categorize the various ways in which software engineering researchers have evaluated the efficacy of LIWC in their studies.

A notable observation among the papers that used LIWC is that the majority, specifically 26 of 43 papers, did not formally evaluate LIWC’s output (see Table 8). These papers relied on LIWC for various purposes and accepted the resulting psychometric measures without any evaluation. In these studies, LIWC’s output was either used for qualitatively analyzing the text or was combined with other data to facilitate subsequent analysis. Some of these studies have used the combination of LIWC’s output and other data points to train machine learning models for tasks such as the prediction of the low quality Stack Overflow posts [S1, S39] or the identification of different arguments within a discussion i.e, argument extraction [S11]. These papers mainly justified their decision based on the established reputation of LIWC as a widely recognized and reliable psychometric text analysis tool, cited by thousands of studies [22].

Sixteen of the papers evaluated LIWC indirectly, meaning that instead of evaluating the output of LIWC itself, they assessed the tools or models

built using LIWC as part of their feature sets. These evaluations often focused on the performance of the models or tools, sometimes incorporating ground-truth data, such as personality questionnaires, to measure accuracy or effectiveness. For example, some studies used LIWC’s categories in machine learning models for tasks like predicting low-quality Stack Overflow posts [S1, S39] or extracting arguments from discussions [S11]. [S38] also combined LIWC’s negative words with other lexicons to create a tool that detects ‘expectation violations’ in mobile app reviews, yielding an accuracy of 96% on a manually labeled subset of 400 data points. In these studies, the focus was on evaluating the overall performance of the models or tools rather than directly assessing LIWC’s individual output.

One study, [S9], performed a comparison of LIWC’s sentiment analysis performance with another model, NLTK VADER, and found them to be perform comparably. However, this was not a formal evaluation of LIWC itself, as it simply compared the performance between the two tools without directly assessing LIWC’s internal accuracy or outputs in isolation. Despite this, we still wanted to distinguish it as a valuable comparison for understanding how LIWC performs relative to other sentiment analysis tools.

Although LIWC has established a strong reputation and is widely used in the scientific community, there remains a need for more formal evaluations of its effectiveness within the software engineering context. While LIWC’s psychometric measures are often trusted without direct evaluation, formally assessing its accuracy and applicability within SE research would benefit the community. However, this is not always a straightforward task. Evaluating LIWC in tasks like sentiment analysis may be relatively simple, but assessing

its measures of psychological concepts, such as personality or emotions, is more complex and requires validation through psychological instruments like specific questionnaires or personality tests. So far, no significant effort has been made to evaluate LIWC directly within software engineering, leaving a gap for future research to address.

RQ4 Summary: Out of the 43 papers examined, 26 did not conduct a formal evaluation of LIWC’s output, relying instead on its established reputation in the scientific community. Of the remaining 17 papers, most assessed LIWC indirectly by evaluating models or tools that incorporated LIWC’s outputs, often using ground-truth data.

5.6. *RQ5: What concerns or limitations were mentioned by the researchers adopting LIWC in the software engineering domain?*

Table 9: LIWC-related Limitations and/or Concerns Expressed in the Selected Studies

Limitation of LIWC	Studies
Domain specificity (e.g., lack or misrepresentation of SE words)	S2, S10, S19, S26, S27, S31, S32, S34, S38
Language limitation (e.g., English lexicon)	S7
Bias against non-native speakers	S27
Lack of qualitative analysis	S21
Limited sentiment analysis capabilities on informal communications	S9
Inability to account for negation	S29

To answer this RQ, we have outlined all the concerns or limitations expressed in the 43 papers we studied. Table 9 presents an overview of how these studies have either acknowledged or overlooked the limitations associated with the use of LIWC. 29 out of the 43 papers refrain to acknowledge potential limitations of using LIWC. In 14 studies, the authors acknowledge

the conceivable limitations of LIWC, delving deeper into the issues or offering potential solutions to mitigate the problems to varying degrees. Some of these studies offer detailed explanations of the concerns pertaining to the adoption of LIWC. We find that these concerns are either inherent limitations of lexicon-based tools or otherwise, related to the application of LIWC to software engineering. Next, we discuss both types of concerns.

5.6.1. Concerns with LIWC’s inherent limitations as a lexicon-based tool

Much like any other lexicon-based tool, the use of LIWC comes with its own set of limitations. Some of these limitations are inherent and often bound by its technical constraints, such as the *inability to account for negation or understanding the context* in a discussion [S29]. For instance, according to LIWC, the sentence “This is not good” contains positive emotion, since the word “good” is associated with positive emotion in LIWC’s dictionary.

Another notable limitation of LIWC is its *limited availability of dictionary languages*, as highlighted by Rigby et al. [S27]. While LIWC2015 includes a German dictionary as its sole non-English option, it’s worth noting that individuals have voluntarily undertaken the considerable task of translating the English LIWC dictionaries (2001, 2007, and 2015 versions) into languages such as simplified and traditional Chinese, Spanish, Italian, Russian, etc. These translated dictionaries are accessible to paid users on LIWC’s official website⁶. The limited number of translated dictionaries may be attributed to the substantial effort required for translation, thus explaining their scarcity. Another language-related limitation of LIWC concerns *text written by non-*

⁶<https://www.liwc.app/dictionaries>

native speakers. Non-native speakers often use a more restricted vocabulary in their communications, potentially influencing the outcomes generated by LIWC [S7].

A potential concern with psycholinguistic tools is that authors might intentionally manipulate their language to influence the analysis. However, this is largely irrelevant in SE contexts, where communications, such as those on GitHub, are organic and not written with the expectation of being psycholinguistically analyzed [S7]. As such, we do not consider this a meaningful limitation for our study.

Overall, while the criticisms aimed at LIWC’s inherent constraints are valid, as it stands, LIWC has been capable of drawing many conclusions from different general-purpose texts successfully [22]. Additionally, as highlighted in the past studies [S29], in practice, when analyzing large corpora of text, the sheer abundance of the data can compensate, to a certain extent, for some of the LIWC’s shortcomings. Therefore, in order to improve LIWC’s performance for software engineering-specific text we might need to turn our attention to the concerns regarding the adoption of this tool in the software engineering domain.

5.6.2. Concerns with LIWC’s adoption in SE

Many studies acknowledge the limitations of adopting LIWC in a software engineering context. Previous research that has identified numerous correlations between various LIWC categories and a wide range of social and psychological phenomena has primarily focused on general text [22]. Given the complexities associated with the transition of psychometric tools to other fields [29], care should be taken when adopting tools such as LIWC in Soft-

ware Engineering studies.

LIWC, may miss many of the words used in particular domains such as software engineering, since *domain specific words may be absent* from its dictionaries. LIWC’s output is as good as its dictionaries and therefore, if the dictionary misses many of the words present in the text, the reliability of LIWC’s output can be severely questioned. It has been observed that some older versions of LIWC can only capture around 66% of the words in software engineering text, discarding words such as API, XML, URI, and HTTP [S2, S31, S34, S28]. However, it is worth mentioning that LIWC’s developers, have been extending its dictionaries on each new release and although the size of a dictionary will always remain limited, LIWC-22 has the largest LIWC dictionary yet, with a size of over 12000 lexicons.

The *misrepresentation of domain-specific words* can also limit the use of LIWC in software engineering research. In many cases, the performance of the tool can be questioned since certain words can have different meanings in particular contexts compared to general text. Certain software engineering-specific terms like “bug”, “kill”, or “cookie” may not be accurately represented in a LIWC dictionary. For instance, the term “cookie,” appears in the food category of the LIWC-22 dictionary. However, in the context of software development, “cookie” typically refers to a type of data transmitted by a website to a user’s web browser.

Additionally, it can be argued that, so far the assessment of LIWC’s direct interpretability for psychological concepts and human factors has been mostly limited to general text. Therefore, to genuinely assess the tool’s reliability for domain-specific text, it is imperative to replicate studies exploring

the correlations between LIWC’s output and psychological concepts using software engineering text such as, developers’ mailing lists and code reviews. Such an approach could help mitigate the performance loss in tasks, such as personality detection, that involve understanding domain-specific communication nuances [S19].

Although our study primarily focused on concerns focused on LIWC’s use in software engineering, some studies have also noted broader ethical considerations when employing psychometric tools in this domain. For example, S19 acknowledges concerns regarding the use of personality detection techniques in SE research. While such concerns are not specific to LIWC itself, the use of automated psycholinguistic tools can raise important questions around privacy, consent, and the interpretation of psychological inferences. We acknowledge the importance of these issues and note that researchers should exercise care when applying such tools to sensitive or personal aspects of developer behavior.

RQ5 Summary: We identify two types of concerns with the use of LIWC in SE: *(a)* inherent limitations due to its lexicon-based nature (e.g., lack of negation handling, limited language support), and *(b)* concerns with domain adoption (e.g., absence and potential misrepresentation of SE-specific words).

6. Discussion

Overall, our study indicates that LIWC has been successfully applied across a broad spectrum of SE tasks involving human factors, such as analyzing team dynamics, detecting emotions, and investigating various BSE con-

cepts (e.g., leadership and personality). Studies have also demonstrated how some psycholinguistic frameworks from other domains that leverage LIWC can be adopted in SE studies, enriching SE research. However, several weaknesses were identified as well. Most notably, few papers rigorously evaluated the validity of LIWC’s outputs for their specific SE contexts; many appeared to assume that the tool’s general reliability would naturally extend to SE text. Furthermore, our findings show that no effort has been made to develop SE-specific dictionaries, which could address domain-related limitations and improve LIWC’s interpretability for SE-specific text. As a result, more validation efforts are needed to ensure that LIWC’s psychological categories can accurately assess SE text.

Addressing these limitations can involve both methodological innovation and integration with emerging technologies. One promising avenue involves combining LIWC with recent advances in natural language processing, particularly Large Language Models (LLMs), which have rapidly gained popularity for text analysis across domains. In recent years, researchers and practitioners have increasingly leveraged LLMs to analyze text; however, unlike LIWC, LLMs operate as black-box models, making their decision-making processes difficult to interpret. While LLMs perform well in generating human-like text and handling various NLP tasks, they often lack transparency when it comes to understanding how specific psychological or emotional dimensions in text are being analyzed. In contrast, LIWC offers a well-established, interpretable framework for extracting psycholinguistic insights, using predefined categories backed by years of validation in psychological research [22]. This allows for clear, consistent analysis of psychological aspects in text,

providing insights that are more directly interpretable and reliable for examining human factors in software engineering. In the following section, we present several future research directions and discuss how LIWC’s psychologically grounded insights can complement and even enhance the capabilities of modern language models.

7. New Research Opportunities Leveraging LIWC

LIWC offers exciting new possibilities that have not yet been extensively explored. Beyond the applications that we listed in Table 5, LIWC can be leveraged for several other tasks, such as:

Evaluating code comments: LIWC can be leveraged to analyze developer-written code comments. By analyzing linguistic features to identify helpful and informative comments, researchers can identify metrics associated with comments that are perceived to be more helpful for the development process.

Predicting project outcomes: Project outcomes, trajectories, and occurrences such as delays or success, can be understood and analyzed through the lens of developer communications. Through analyzing textual communications of developers e.g., mailing lists, code reviews, GitHub discussions, etc. researchers can further analyze the human aspects of development and their interplay with project outcomes. Analyzing linguistic patterns in developer communications can reveal certain patterns of communication that are frequently associated with positive or negative outcomes.

Identifying developer burnout: By monitoring developer communication in project management tools and version control systems, future studies can aim to detect signs of burnout or emotional stress among team members

using categories such as *Stress*, *Social processes*, *Assent*, etc.

Enhancing software documentation: Identifying areas of improvements such as readability or ambiguity within project documentations can help address some of the relevant concerns. Further such analyses can help developers identify whether certain type of language or presentation is more commonly perceived to be helpful.

Improving product or app reviews: LIWC can help in detecting linguistic patterns that reflect user satisfaction or indicate areas for improvement of software.

Software security and social engineering: Additionally, similar to the work by Janjua et al. [S41], LIWC can also be used to further our understanding of social engineering-based attacks in software projects by analyzing the communications between attackers and project members. While [S41] used LIWC categories as feature sets for models, a more direct linguistic analysis, focusing on the specific language patterns employed by attackers, can offer deeper insights into the behaviors and strategies behind social engineering attacks. This approach could help identify linguistic markers of manipulation, persuasion, and trust-building tactics in the context of these attacks.

Software engineering specific dictionaries: As detailed in Section 5.6.1 certain concerns associated with LIWC are inherent in any lexicon-based approach while others are the result of adopting the tool in a new field. Among the latter concerns, the issue often brought up in the literature is the domain specificity of software engineering text and LIWC’s limitations in capturing the text’s context. The answer to this problem may lie in the

creation of software-specific LIWC dictionaries which can prove to be a huge step forward toward a better understanding of developers and their unique linguistic patterns. While creating any LIWC dictionary demands a notable investment of time and effort, once created, it can present new opportunities for those studying developers' communications. Such dictionaries can be specific to particular settings in order to capture the language patterns in certain situations. For example, researchers interested in investigating the levels of stress and anxiety among developers could develop a custom dictionary that reflects the language use of developers in high-pressure situations. Furthermore, custom dictionaries could be defined with specific software engineering constructs, such as Team Management or Quality Assurance, in mind to more accurately capture the language patterns and psychological markers relevant to each activity. For example, dictionaries designed around team coordination could emphasize constructs like leadership, trust, or conflict resolution, enhancing the interpretability of LIWC's output in those contexts. Overall, LIWC possesses the potential to significantly improve our understanding of the text written by developers and therefore, addressing some of LIWC's limitations can prove to be invaluable.

Leveraging more LIWC categories for direct analysis of the text:

Another way to improve the usability of LIWC is to take advantage of all its categories for direct analytical purposes. LIWC presents an expansive set of word categories. However, commonly, studies select specific categories, tailored to the needs of their particular analytical objectives, for close examination. As evident in Table 1, our findings point to the diversity of LIWC categories that have been utilized to serve different research purposes

in software engineering studies. Yet, even though many software engineering studies have leveraged several of LIWC’s capabilities, we still believe that the program holds untapped potential for conducting further text analysis through direct interpretation of its categories. Consider the category of “assent”, known to correlate with agreement and passivity [22]. Understanding assent can potentially help us in studying team conflicts, or analyzing discussions involving critical decision-making, such as tool adoption. Despite its potential relevance, this category has yet to be employed in order to assess agreement levels within developer conversations.

Improving the performance of learning-based models: One promising approach involves integrating LIWC to enhance their ability to analyze text. As highlighted in our response to RQ1, LIWC categories have already been utilized as input in various machine learning models, often in combination with other features such as metadata from sources like Stack Overflow posts or GitHub repositories.

LIWC features could even be incorporated into ensemble classifiers to improve performance across SE-related prediction tasks. By leveraging LIWC’s psychologically grounded insights, deep learning models could also be enriched with valuable context that improves their ability to capture nuanced human emotions and cognitive states. Given the challenges in interpretability that many deep learning models face [90], this hybrid approach offers a promising direction for future research and application.

Analyzing LLM-generated text: Additionally, there is considerable room for analyzing LLM-generated content with the help of LIWC. Beyond evaluating the content of ChatGPT-generated responses, Kabir et al. an-

alyzed the linguistic characteristics of such content generated in response to Stack Overflow questions and compared them to those written by humans [S43]. Their findings revealed that LLM-generated content tends to be more formal, analytic, and positive in language. These distinctions also suggest the potential for using LIWC to help differentiate between human- and LLM-authored text in software engineering artifacts. This kind of analysis can be extended to explore the linguistic features of different LLMs in various scenarios and conversations in software contexts. For instance, it would be insightful to determine what type of language is most effective in prompting an LLM when trying to perform programming tasks or address GitHub issues in a repository. Furthermore, it would be valuable to study how LLMs navigate conversations based on specific linguistic cues, whether they adjust their style in certain contexts, and how these adjustments impact the usefulness of their responses.

Moreover, LLMs are susceptible to inherent biases in the text they generate, largely due to the vast, unfiltered datasets on which they are trained. LIWC can be instrumental in identifying such biases by analyzing language patterns in LLM-generated texts, particularly for sensitive categories related to gender, emotions, or ethnicity. For example, studies outside the software engineering domain have demonstrated gender differences in how LLMs replicate lexical, psychological, and social traits [91]. Applying these analyses to software engineering could reveal whether biases related to developers' identities emerge in LLM-generated responses in software engineering contexts e.g., addressing programming issues or performing tasks such as bug fixes or vulnerability detection. This could lead to important insights on how LLMs

contribute to or mitigate biases in collaborative development environments.

8. Threats to Validity

Construct Validity: Papers retrieved from online libraries were manually filtered to select those pertinent to this study. While this method is susceptible to human error, we adopted a two-step process: one author conducted the primary selection, while another independently verified the final list. Although this adds a layer of validation, it does not fully replicate having two authors review every paper, and some risk of missing relevant studies remains. In terms of our search query, a potential threat to validity is the assumption that the naming conventions for LIWC are consistent across all relevant literature, which may overlook studies using alternate terminology. We mitigated this threat by explicitly checking for alternate naming conventions during our study but found none in the relevant studies across software engineering, psychology, and communication domains. In addition to assessing the title and abstract, we thoroughly examine the entire paper before determining its inclusion in our study. This approach ensures the precision and relevance of our paper selection. To further mitigate bias, two authors were involved in the data extraction and coding phases of the study. We followed an iterative discussion and coding procedure until we reached substantial agreement for each category (above 0.6).

Internal Validity: The internal operations of the online libraries we utilized in our study are undisclosed. Their methods of indexing studies and responding to search queries are not publicly available. Therefore it is possible, that this search process has potentially resulted in the omission of

relevant studies. However, to mitigate this threat, we have collected relevant papers from six prominent and widely recognized online digital libraries, i.e., IEEE Xplore Digital Library, ACM Digital Library, Elsevier Science Direct, Springer Link Online Library, Wiley Online Library, and Scopus.

All included papers were peer-reviewed publications in reputable venues, which we believe minimizes bias risk. For this reason, we did not conduct a formal risk of bias assessment, though we acknowledge this as a potential limitation of our study.

External Validity: In order to narrow down the survey to relevant papers, we implemented stringent inclusion and exclusion criteria. This led to our focus on papers written in English and accessible through one of our six chosen databases. However, this approach may have missed some papers, potentially affecting the study’s generalizability. To mitigate this risk, we designed our search string to be as inclusive as possible, deliberately introducing more noise into our initial results. We later performed a meticulous manual analysis to eliminate irrelevant entries. This strategy aimed to ensure the inclusion of as many relevant papers as possible. Consequently, we invested significant time in the manual inspection process, resulting in the removal of over a thousand irrelevant papers.

9. Conclusion

The primary objective of this paper is to investigate the role of LIWC, a representative psycholinguistic tool widely recognized for its effectiveness in analyzing language and psychological patterns, in the software engineering research. Through this study, we aim to identify LIWC’s utility, outline its

limitations and potential in SE research and practice. The selection of the 43 papers took place through a systematic process, using specific inclusion and exclusion criteria, as discussed Section 4. The findings from our five research questions, as summarized next, can serve as a potential resource for future work in the field, looking to utilize LIWC.

In addressing our first research question, we underlined the diverse purposes for which LIWC has been used within the software engineering domain. Our findings reveal an almost even split between studies that directly interpret LIWC’s various dimensions and those that employ it in an indirect manner, often combined with other data or as input for machine learning models. We also highlight how, throughout the years, the community has demonstrated many unique methods of leveraging LIWC to address development-related issues and enhance the software engineering process. Furthermore, we present the types of data and data sources that were utilized in our selected studies, and leverage a taxonomy of software engineering activities adopted from previous research, to identify the activities that were most frequently studied using LIWC (e.g., *Team Management* and *Knowledge Management*). Additionally, by adopting Behavioral Software Engineering (BSE) concepts, we carefully annotated each paper based on the concepts it examined, offering a precise overview of the distribution of these studied concepts. This step allowed us to identify the concepts that were frequently studied using LIWC, as well as the ones that have received less attention or remained unexplored. Next, to better understand the potential and limitation of LIWC in analyzing domain-specific language used by software engineering practitioners, we investigated the evaluation methods used by researchers. Surprisingly, we

discovered that a substantial portion of the papers neither directly evaluated LIWC’s findings nor raised any concerns regarding its usage. Nonetheless, a noteworthy number of papers do subject LIWC’s findings to different types of evaluations. We also note that ethical considerations, such as privacy and the potential for misinterpretation of psychometric inferences, are rarely addressed.

In summary, this study’s findings demonstrate LIWC’s unique ability to capture linguistic patterns that enhance our understanding of the human factors impacting software development and maintenance. While we believe that more of LIWC’s capabilities remain to be explored, it is essential we understand the limitations of using an off-the-shelf textual analysis tool for SE-specific text. This study aims to shed light on the significance, applications, implications, and potential use of LIWC in software engineering research. We hope that the insights from this work will empower future research to better recognize and leverage the possibilities offered by LIWC.

Appendix A. Selected Studies

Below you can find a list of the studies used in this SLR:

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[S2] Licorish, S. A. and MacDonell, S. G. (2014a) ‘Personality profiles of

global software developers’, in Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering. London England United Kingdom: ACM, pp. 1–10. doi: 10.1145/2601248.2601265.

[S3] Calefato, F. et al. (2018) ‘Sentiment Polarity Detection for Software Development’, *Empirical Software Engineering*, 23(3), pp. 1352–1382. doi: 10.1007/s10664-017-9546-9.

[S4] Wang, Y. and Redmiles, D. (2016) ‘The Diffusion of Trust and Cooperation in Teams with Individuals’ Variations on Baseline Trust’, in Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing. San Francisco California USA: ACM, pp. 303–318. doi: 10.1145/2818048.2820064.

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Software Reliability Engineering (ISSRE). Ottawa, ON, Canada: IEEE, pp. 77–86. doi: 10.1109/ISSRE.2016.43.

[S8] Calefato, F. and Lanubile, F. (2022) ‘Using Personality Detection Tools for Software Engineering Research: How Far Can We Go?’, *ACM Transactions on Software Engineering and Methodology*, 31(3), pp. 1–48. doi: 10.1145/3491039.

[S9] Hergueux, J. and Kessler, S. (2022) ‘Follow the Leader: Technical and Inspirational Leadership in Open Source Software’, in *CHI Conference on Human Factors in Computing Systems*. New Orleans LA USA: ACM, pp. 1–15. doi: 10.1145/3491102.3517516.

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[S11] Wang, W. et al. (2020) ‘ArguLens: Anatomy of Community Opinions On Usability Issues Using Argumentation Models’, in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Honolulu HI USA: ACM, pp. 1–14. doi: 10.1145/3313831.3376218.

[S12] Mantyla, M. V. et al. (2017) ‘Bootstrapping a Lexicon for Emotional Arousal in Software Engineering’, in *2017 IEEE/ACM 14th Interna-*

tional Conference on Mining Software Repositories (MSR). Buenos Aires: IEEE, pp. 198–202. doi: 10.1109/MSR.2017.47.

[S13] Gachechiladze, D. et al. (2017) ‘Anger and Its Direction in Collaborative Software Development’, in 2017 IEEE/ACM 39th International Conference on Software Engineering: New Ideas and Emerging Technologies Results Track (ICSE-NIER). Buenos Aires: IEEE, pp. 11–14. doi: 10.1109/ICSE-NIER.2017.18.

[S14] Raman, N. et al. (2020) ‘Stress and burnout in open source: toward finding, understanding, and mitigating unhealthy interactions’, in Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering: New Ideas and Emerging Results. Seoul South Korea: ACM, pp. 57–60. doi: 10.1145/3377816.3381732.

[S15] Keertipati, S., Savarimuthu, B. T. R. and Licorish, S. A. (2016) ‘Approaches for prioritizing feature improvements extracted from app reviews’, in Proceedings of the 20th International Conference on Evaluation and Assessment in Software Engineering. Limerick Ireland: ACM, pp. 1–6. doi: 10.1145/2915970.2916003.

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Lisbon, Portugal: IEEE, pp. 18–23.

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[S24] Han, Y. et al. (2023) ‘Cross-status communication and project outcomes in OSS development: A language style matching perspective’, Empirical Software Engineering, 28(3), p. 78. doi: 10.1007/s10664-023-10298-8.

[S25] Júnior, M. C. et al. (2021) ‘Industrial and OSS developers’ profiles: a family of experiments to evaluate a pioneering neuro-linguistic method for preferred representational systems automatic detection’, Journal of the Brazilian Computer Society, 27(1), p. 4. doi: 10.1186/s13173-021-00107-9.

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pp. 1578–1596. doi: 10.1016/j.infsof.2014.02.004.

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[S39] Xia, X. et al. (2016) ‘It takes two to tango: Deleted stack overflow question prediction with text and meta features’, in 2016 IEEE 40th annual computer software and applications conference (COMPSAC). IEEE, pp. 73–82.

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[S41] Janjua, F., et al. (2020). ‘Handling insider threat through super-

vised machine learning techniques’. In *Procedia Computer Science*, 177, pp. 64–71. doi: 10.1016/j.procs.2020.10.012

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Appendix B. PRISMA 2020 Items

Table B.10: PRISMA 2020 Items with Reporting Locations

Checklist Item	Location Reported
Title: Identify the report as a systematic review.	Title
Abstract: See the PRISMA 2020 for Abstracts checklist.	Abstract
Introduction – Rationale: Describe the rationale for the review in the context of existing knowledge.	Section 1
Introduction – Objectives: Provide an explicit statement of the objectives or questions the review addresses.	Section 1
Methods – Eligibility criteria: Specify inclusion and exclusion criteria and how studies were grouped.	Section 4.2
Methods – Information sources: Specify all sources searched and date of last search.	Sections 4 & 4.2

Checklist Item	Location Reported
Methods – Search strategy: Present full search strategies for all sources.	Section 4
Methods – Selection process: Describe methods for deciding study inclusion and number of reviewers involved.	Section 4
Methods – Data collection process: Specify methods used to collect data.	Section 4.3
Methods – Data items (Outcomes): List and define all outcomes for which data were sought.	Section 4
Methods – Data items (Other variables): List and define all other variables for which data were sought.	N/A
Methods – Study risk of bias assessment: Methods used to assess risk of bias.	Section 4
Methods – Effect measures: Effect measures used for each outcome.	N/A
Methods – Synthesis methods: Describe methods used to synthesize results and explore heterogeneity.	Section 4
Methods – Reporting bias assessment: Describe methods used to assess risk of bias due to missing results.	N/A
Results – Study selection: Describe results of search and selection.	Section 5
Results – Excluded studies: Cite studies that may appear to meet inclusion criteria but were excluded.	Supplementary Material
Results – Study characteristics: Cite each included study and present characteristics.	Appendix A and Section 5
Results – Results of individual studies: Present summary statistics and effect estimates.	N/A
Results – Results of syntheses: Present results of all statistical syntheses, including heterogeneity.	N/A
Discussion – Summary of evidence: General interpretation of the results in the context of other evidence.	Section 6
Discussion – Limitations of evidence: Discuss limitations of included studies.	Section 6
Discussion – Limitations of review process: Discuss limitations of the review process.	8
Discussion – Implications: Discuss implications for practice, policy, and future research.	Section 6

Checklist Item	Location Reported
Other – Availability of data: Report availability of data, code, and materials.	Section 4.3

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